

STATEMENT OF WORK

FY26 CYCLIC ROADWAY STRIPING

Blue Ridge Parkway
Augusta, Nelson, Amherst, Rockbridge, Bedford, Botetourt Counties, Virginia



PART 1 – GENERAL

1-1 DESCRIPTION OF WORK

- A. Heavy traffic conditions, created by over 16 million annual visitors to the Blue Ridge Parkway (parkway), along with snow plowing operations and extreme winter freezing conditions, have contributed to the accelerated deterioration of roadway striping. The work consists of installing centerline striping in accordance with these specifications along sections of the parkway in North Carolina.
- B. This Statement of Work (SOW) consists of matching existing centerline striping along designated sections of the parkway. Required work shall be performed along northern sections of the Blue Ridge Parkway.

REQUIRED WORK					
Section	MP Start	MP End	Estimated Distance (MI)	Estimated Striping (LF)	Notes:
Access Ramp to U.S. Highway 250	-	0.10	0.19	2,006	At Mile Post 0.10
0001AA (First Part)	0.00	27.25	27.25	287,760	North End: From end of Sky Line Drive (North Side of I-64 Bridge) South End: Intersection with VA Highway 56 / Tye River Turnpike
Access Ramp to VA State Highway 56	-	27.25	0.08	845	At Mile Post 27.25 on Left

- C. Option work consists of matching existing centerline striping along designated sections of the parkway. All, some, or none of the option work may be included in the project as authorized by the CO.

OPTION 1					
Section	MP Start	MP End	Estimated Distance (MI)	Estimated Striping (LF)	Notes:
0001AA (Second Part)	27.25	37.6	10.35	109,296	North End: Intersection with VA Highway 56 / Tye River Turnpike ramp South End: Irish Gap, underpass VA State Road 805 (Pedlar River Rd / Nettle Creek Rd)

OPTION 2					
Section	MP Start	MP End	Estimated Distance (MI)	Estimated Striping (LF)	Notes:
0001AB (First Part)	37.6	45.62	8.02	84,691	North End: Irish Gap, underpass VA State Road 805 (Pedlar River Rd / Nettle Creek Rd) South End: Intersection with U.S. Highway 60, Lexington Turnpike ramp
Access Ramp to U.S. Highway 60	-	45.62	0.19	2,006	At Mile Post 45.62 on Left

OPTION 3					
Section	MP Start	MP End	Estimated Distance (MI)	Estimated Striping (LF)	Notes:
0001AB (Second Part)	45.62	63.60	17.98	189,869	North End: Intersection with U.S. Highway 60, Lexington Turnpike ramp South End: North end of James River Bridge (not including the bridge)
Access Ramp to VA State Highway 130	-	61.58	0.04	422	At Mile Post 61.58 on Left
Access Ramp to U.S. Highway 501 / Lee Highway	-	64.13	0.20	2,112	At Mile Post 64.13 on Left

OPTION 4					
Section	MP Start	MP End	Estimated Distance (MI)	Estimated Striping (LF)	Notes:
0001AB (Third Part)	63.9	74.89	10.99	116,054	North End: South end of James River Bridge (not including the bridge) South End: Where the Appalachian Trail crosses the Blue Ridge Parkway

D. Attachments:

1. CM-16 Transmittals Form (Submittals)
2. BLUE RIDGE PARKWAY LOAD LIMITS
3. SUGGESTED OUTLINE FOR CONTRACTOR'S SAFETY PLAN

1-2 LOCATION AND ACCESS

A. Job Site: The work shall be performed on various sections of the parkway in Virginia. The mileposts are provided in SOW Sections 1-1.B and 1-1.C. The work is being performed in the following counties: Augusta, Nelson, Amherst, Rockbridge, Bedford, Botetourt.

B. Access

1. There are many ways to access the parkway sections of this contract. Use either independent navigation resources or the Blue Ridge Parkway Real-time Road Closure Map to navigate to the various parkway sections of this contract. The Blue Ridge Parkway Real-time Road Closure Map is available for free here: [Road Status and Closures - Blue Ridge Parkway \(U.S. National Park Service\)](https://www.nps.gov/blri/planyourvisit/roadclosures.htm) (<https://www.nps.gov/blri/planyourvisit/roadclosures.htm>).
2. The maximum permitted load limits for Parkway roads accessing the work site are described in attachment BLUE RIDGE PARKWAY LOAD LIMITS.
3. Special consideration may be made by the Contracting Officer in consultation with the park engineer for overweight and oversize loads. Special permits, times and traffic control may be required for heavy/wide loads.
4. All damage shall be repaired to the Government's specification by those who cause it and at no cost to the Government. Return the area to the condition existing prior to any work.

1-3 PROJECT PERSONNEL

A. Government

1. The Contracting Officer (CO) is the only person with authority to enter into, administer, modify and/or terminate the contract.
2. The Contracting Officer's Representative (COR) is a program official designated by the CO to assist in administering technical aspects of the contract. The COR cannot modify the contract.
3. Other NPS personnel may be on site to assist with project inspection. These individuals cannot modify the contract. Any on site issues or concerns shall be directed to the COR.

B. Contractor

1. Assign a competent and experienced person to oversee the construction and safety of this project with the authority to stop or correct work due to safety concerns, unsatisfactory materials, or workmanship.
2. Use adequate numbers of skilled workers who are fully qualified and experienced in the type of work to be executed in this contract. When applicable, personnel shall be certified in the necessary work for which being performed by that person. The COR may at any time request copies of necessary certifications.
3. Traffic Control Personnel
 - a. Provide onsite at all times a traffic control supervisor to oversee work zone operations, installation, and maintenance of temporary traffic control devices. The individual(s) shall be certified by a state department of transportation, American Traffic Safety Services

Association (ATSSA), or other acceptable certification programs with a minimum of two years of Temporary Work Zone Traffic Control "On the Job Training" (work involving the setup, take-down and or inspection of Work Zones within existing Highway Right-of-Way). The traffic control supervisor shall regularly oversee the work zone temporary traffic control but may perform minor rolls in the work.

- b. Provide flaggers, spotters, or operators of Automated Flagger Assist Devices (AFAD) certified by a state department of transportation, ATSSA, or other acceptable certification programs.
- c. Prior to work beginning, provide contact information and certification(s) of person(s) responsible for the individuals meeting SOW Section 1-3.B.3.a and b.
- d. Individuals meeting the above requirements shall carry a card issued by the certifying firm that shows the dates of certification with recertification at regular intervals.
- e. See SOW Section 3-3.

1-4 CONTRACTOR-FURNISHED ITEMS

- A. The contractor shall furnish all necessary supervision, personnel, equipment, tools, transportation, supplies, materials, and incidentals required to perform the work of the contract in a safe, professional manner and to satisfactorily complete the work of this project.
- B. The contractor shall furnish all necessary temporary traffic control and signage to alert motorists of work in progress and to protect the work from damage.
- C. Work site security measures. Nearly all Parkway land is accessible by the public 24-hours a day throughout the year. Parkway law enforcement officers and other staff will not be designated to provide extra security to protect project sites from visitor intrusion. It is the responsibility of the Contractor to take measures to prevent vandalism, theft, or destruction of project materials and equipment. Any proposed security measures that may affect ongoing access or use of areas otherwise not already included in the construction zone (including remote security cameras or alarms) must be discussed with and approved in advance by the COR, in consultation with the parkway District Ranger. Any security issues must be reported to the COR immediately upon notice. Parkway law enforcement staff may request official statements or other information from Contractor personnel needed to follow up on reported issues.
- D. Equipment used to install the roadway striping shall have the following characteristics:
 - 1. Truck-mounted and self-contained vehicle. The vehicle shall be in good working order, routinely used in the type of work for this contract and have current the necessary certifications and inspections as required by the state in which it is registered.
 - 2. Pressurized paint nozzles located outboard of the vehicle wheels.
 - 3. On-off control for bead dispenser synchronized with the On-Off control for the paint nozzle.
 - 4. Provide a pressurized bead dispenser/applicator.
 - 5. Must have a measuring device capable of automatically and accurately measuring and totaling the stripe lengths.
 - a. Accuracy shall be maintained, via field calibration, within 1 percent plus or minus of actual measured length.
 - b. If equipment does not have field calibration capability, any error in excess of 1 percent shall be mathematically corrected for payment.

6. A heating device capable of heating and delivering paint to the spray nozzle at a minimum of 140 degrees Fahrenheit or as recommended by the paint manufacturer to improve quick dry characteristics.
7. Capable of applying hot paint under pressure directly to the pavement with a uniform width and thickness to produce clean, sharp, and smooth edges on stripes.
8. Capable of automatically or manually applying two parallel solid or broken stripes simultaneously.
9. The spray nozzle and bead dispenser shall be shielded to prevent over-spraying.
10. Mobile and maneuverable to the extent that straight lines can be followed, and all standard curves can be made in true arcs.

1-5 WORK CONDITIONS AND LIMITATIONS

A. Facility Use

1. The parkway motor road is open to visitors year-round, 24-hours a day weather permitting. Visitation is highest during summer months and especially leaf viewing season (typically October).
2. These sections of parkway provide access to several popular destinations.

B. Required Performance Period

1. The performance period for this project is defined in the contract documents.
2. No work is permitted during the month of October.
3. No work will be permitted at night.
4. No work will be permitted on Federal holidays.
5. No work will be permitted on weekends – Friday 1:00 p.m. to Monday at 6:00 a.m.
6. All work shall be complete, equipment demobilized, excess materials removed, and the work area(s) ready for visitor use by the last day of the performance period.
7. For any temporary periods of no work performance, equipment and materials shall be confined to areas designated by the COR. The roadway shall be open and safely passable for visitor use. Temporary traffic control shall be covered or removed during this period if not applicable.
8. Roadway striping in the tunnels are a different material and not included in this contract. No roadway striping is to be installed inside any tunnels as part of this contract.

C. Contractor's Use of Premises

1. Due to limited available storage space at parkway facilities, the Contractor shall arrange to store all paint, beads, and any other materials and equipment off Government land.
2. Staging and parking for vehicles, equipment, and materials shall be confined to paved areas and areas specifically designated by the COR. For other areas, provide written request to the COR three days prior to occupying these other areas.
3. The only exception shall be overnight parking of the paint striping truck at parkway facilities on a space available basis. Request overnight parking in writing to the CO five days in advance.
4. All project materials shall be delivered and stored in the manufacturer's original suitable container and remain in such containers until immediate use. Material packaging shall be clearly and adequately marked with the material, color, date of manufacture, process, batch or lot number, manufacturer's name and location, temperature range, shelf life, and include the material safety data sheets (MSDS).

5. Keep work site clear of trash and debris. Dispose of any trash, debris, and other unusable material legally at authorized facilities on non-Federal land.
6. Food trash and debris shall be placed in the appropriate receptacles at the end of each day.
7. Restrooms along the parkway may be closed during the performance period.

D. Existing Utilities

1. No utilities are expected to be impacted by this project.
2. If utilities are encountered, cease work and notify the COR immediately.

E. Field Verification

1. The measurements and quantities specified are estimated and provided for information only. The Contractor shall verify all dimensions and conditions affecting the work.
2. The Contractor shall notify the COR of any errors or omissions as soon as realized.

F. Park-Specific Conditions and Limitations

1. The Blue Ridge Parkway is wholly located along the ridges of the Appalachian Mountain range in parts of Virginia and North Carolina. The following conditions may be applicable to work sites inside Parkway boundaries:
 - a. WEATHER- Sudden and dramatic changes in climate conditions are common. The Contractor must be prepared to alter work plans and provide protection for workers and equipment.
 - b. CLOSURES- Parkway officials may close portions of the Parkway at any time to protect visitors from hazardous conditions or to implement official duties of the Parkway.
 - c. REMOTENESS- Many portions of the Parkway are of extreme distance to typical services available in developed areas, including cellular, food, fuel, and emergency services. The Contractor must arrange and account for appropriate precautions and plans prior to bid submission.
 - d. RESOURCES- As a unit of the National Park System, the Blue Ridge Parkway is tasked to preserve unimpaired its natural and cultural resources and values. The Contractor must make every effort to ensure no impacts are made deleterious to these resources. It is expected that the project site and access areas will be minimally impacted only as absolutely necessary for the performance of the SOW requirements. Mitigations approved by the COR will be taken to restore pre-existing conditions.
 - e. ROCK- The presence of various forms of rock is ubiquitous. The Parkway does not maintain data for construction purposes on the location or type of rock to be encountered by excavation or drilling activities. The contractor must verify and account for the extent, nature and impact to work activities of rock within the project site prior bid submission.
 - f. INVASIVE EXOTICS- To reduce the chance that harmful, non-native plants or organisms will be transported from off-parkway locations to the work site, the contractor, prior to entering the parkway, must clean all soils, roots, seeds, and other sources of plant parts and pathogens from any equipment used for the application of grasses and all heavy equipment, such as dump trucks, front end loader, and graders, which have previously been operating off the park.
 - g. WILDLIFE- All wildlife is protected within the park boundaries, and some species, which occur within the park, receive special State and Federal protection due to their rare, threatened or endangered status. Very few wildlife species present any danger to persons working within the park. Even species which have the potential to do harm to persons, can

usually be avoided. Unless wildlife is presenting an immediate threat to persons, harassment or destruction of wildlife is prohibited.

2. Project-specific environmental compliance mitigations
 - a. All debris shall be removed from the park for legal and proper disposal.
 - b. Staging and parking for vehicles, equipment, and materials shall be confined to designated areas approved by the COR.
 - c. All construction activities shall be performed in a manner having the least possible detrimental effect to the parkway's cultural and natural resources.

1-6 QUALITY CONTROL

- A. It is the responsibility of the Contractor to develop and implement a construction quality process to ensure work conforms to the requirements of the contract.
- B. Assign a competent and experienced person to oversee the construction of this project with the authority to stop or correct work due to unsatisfactory materials or workmanship.
- C. Perform quality control inspection, documentation, and process control sampling and testing as required by applicable standards.
- D. All materials and methods used shall meet or exceed the current local, state, and federal codes, standards, and regulations applicable to this project.
- E. The COR will perform on-site inspections to insure conformance to this contract. The Government reserves the right to perform any testing it sees fit to ensure a satisfactory product is delivered. Any non-conforming work will be removed and replaced at no cost to the Government.

1-7 REFERENCE STANDARDS:

- A. Perform all work in accordance with the SOW, Project Plans, and following referenced specifications.
- B. Federal Highway Administration (FHWA) Standard Specifications for the Construction of Roads and Bridges on Federal Highway Projects (FP24). These specifications are available for free download at highways.dot.gov/federal-lands/specs/fp-24.pdf.
- C. North Carolina Department of Transportation (NCDOT) 2024 Standard Specifications for Roads and Structures.
- D. Virginia Department of Transportation (VDOT) 2020 Road and Bridge Specification Book. These specifications are available for free download at: [VDOT 2016 RB Specs.pdf \(viriniadot.org\)](https://www.viriniadot.org/business/materials-download-docs.asp)
- E. Materials shall meet FP-24 specifications or be currently approved for use by NCDOT or VDOT on the approved products list.
 - NCDOT - <https://apps.ncdot.gov/vendor/ApprovedProducts/Default.aspx>
 - VDOT - <http://www.viriniadot.org/business/materials-download-docs.asp>

F. Temporary Traffic Control Devices shall be placed and maintained in accordance with the current version of the Manual on Uniform Traffic Control Devices (MUTCD) and the Project Plans. The MUTCD is available for free download at: <https://mutcd.fhwa.dot.gov/index.htm>.

PART 2 – SUBMITTAL REQUIREMENTS

2-1 GENERAL

- A. Refer to attached CM-16 Transmittal Form for required submittal items. Contractor is responsible for additional submittals that may be required. Electronic forms are available.
- B. Unless otherwise instructed by the CO, send construction material and other items requiring approval to the CO and copy the COR. Provide one (1) electronic copy of each submittal unless otherwise specified. Forward related materials as a package. Reference the SOW section number on each submittal. Allow 7 days for review and approval.
- C. The CO will indicate in writing the approval or disapproval of each submittal, the reason for any disapproval, and any resubmission requirements. One (1) of each approved submittal will be stamped and returned to the Contractor unless otherwise requested.
- D. Until the Contractor receives approval from the CO, no construction material or other items requiring the CO's approval may be ordered, delivered, or incorporated into the project work.
- E. There will be no waiver of any required submittal unless authorized by the CO. The CO may request submittals other than those specified in this document.

2-2 Work Plan Submittals

- A. Project Schedule: Contractor shall submit a construction schedule outlining each phase of the work along with a written narrative of the activities displayed in the schedule.
- B. Safety and Traffic Control Plan
 - 1. Contractor shall submit a site-specific Safety and Traffic Control Plan for the work prescribed in this contract.
 - 2. Refer to the following FHWA publications for further guidance when developing the traffic control component of this plan:
https://www.workzonesafety.org/files/documents/training/courses_programs/rsa_program/RSP_Guidance_Documents_Download/RSP_ITCP_Guidance_Download.pdf
 - 3. The plan shall designate a competent individual capable of overseeing and understanding worker safety and the hazards associated with this type of work to implement the plan.
 - 4. The plan shall describe the potential hazards of the work site for each phase, along with all company policies, controls, work practices selected to minimize those hazards and detail the implementation of the plan.
 - 5. The plan shall contain at a minimum company policy or goals statement, list of responsible persons, hazard identification, hazard controls and safe practices, emergency and accident response, employee training and communication as well as recordkeeping procedures.
 - 6. Provide plans for placing and maintaining both traffic and pedestrian control devices.

7. See attached SUGGESTED OUTLINE FOR CONTRACTOR'S SAFETY PLAN for assistance in developing the site-specific Safety Plan.
8. Refer to the following Occupational Safety and Health Administration (OSHA) website for further assistance: <https://www.osha.gov/Publications/OSHA3886.pdf>

C. Paint Spill Cleanup Plan

1. Contractor shall submit a detailed plan listing the procedures, equipment, and materials to be used to clean up any paint spills that might occur within the parkway during the course of this project.
2. The plan should anticipate cleanup methods not only for spills on asphalt pavement but also on grassed shoulders or adjacent natural ground landscapes.

2-3 MATERIAL SUBMITTALS

A. Submittals: Materials

1. Centerline Roadway Striping (Pavement Marking Paint)
 - a. FHWA Type C – High-build waterborne traffic paint with beads (reflective)
 - b. Furnish a yellow, acrylic water-based, ready-mixed paint conforming to the following:

1.	Composition. Furnish a paint composed of resin solids of 100 percent acrylic polymer with the exact formulation determined by the manufacturer. Conform to the following:	
	a. Pigment, ASTM D3723	45 to 55 percent by mass
	b. Lead, chromium, cadmium, or cobalt ASTM D3335 & ASTM D3718	0 percent by mass
	c. Volatile organic compounds, EPA Method 24	20.0 oz/gallon (150 g/L) max.
	d. Density of paint, ASTM D1475	12.0 lb/gallon (1.44 kg/L) min
2.	Viscosity. ASTM D562	75-90 Krebs units
3.	Drying time	
	a. Dry to no pickup, ASTM D711	10 minutes max
	b. Drying to no track, ASTM D713	90 seconds max
4.	Flexibility. ASTM D522, using the ¼-in (6.4-mm) cylindrical mandrel	No cracking or flaking
5.	Dry opacity. ASTM D2805, contrast ratio at 319 ft ² /gallon (7.85 m ² /L) spreading rate	0.96 min
6.	Color	
	a. White, ASTM D1729	Match FHWA standard highway white
	b. Yellow, ASTM D1729	Match FHWA standard highway yellow
7.	Daylight reflectance. (Without glass beads)	
	a. White, ASTM E1347	84 percent relative to magnesium oxide standard
	b. Yellow, ASTM E1347	55 percent relative to magnesium oxide standard
8.	Bleeding ratio. ASTM D868. Determine reflectance according to ASTM D1347 immediately after drying. Divide the average of three reflectance	0.96 min

readings of the paint over the bleeding surface by the average of three readings over the non-bleeding surface to determine the bleeding ratio	
9. Freeze-thaw stability. ASTM D2243	±5 Krebs units max
10. Storage stability. During a 12-month storage period, conform to the following:	
a. No excessive setting, caking, or increase in viscosity; and	
b. Readily stirred to a consistency for use in the striping equipment.	

2. Glass Beads

- a. Furnish glass beads conforming AASHTO M247, Type I and Type III with the following specific properties:

1.	Gradation			
	Sieve Designation		Mass % Passing	
	Standard, mm	Alternate No.	Type 1	Type 3
	1.70	12		100
	1.40	14		95-100
	1.18	16	100	80-95
	1.00	18		10-40
	0.850	20	95-100	0-5
	0.710	25		0-2
	0.600	30	75-95	
	0.425	40		
	0.300	50	15-35	
	0.180	80		
	0.150	100	0-5	
2.	Roundness. The glass beads shall have a minimum of 70 percent true spheres.			
3.	Refractive Index. The glass beads shall have a refractive index of 1.50–1.55.			
4.	Coatings: Flotation, Flow, Moisture Resistance, and Adherence – When tested in accordance with Sections 8 through 14 of T 346, the presence of the purchaser-specified coatings shall be unambiguous. Beads shall flow properly when tested in accordance with the procedure in Section 8 of T 346 unless specified to be moisture resistant, in which case the beads shall flow properly when tested in accordance with the procedure in Section 9 of T 346.			
5.	General. The beads shall be dry and free of lumps and clusters.			

- b. Treat glass beads with an adherence coating as recommended by manufacturer.

3. Paint Removal Material

- a. Contractor shall provide an industry standard safe and effective product or means for removal and cleanup of spilled or incorrectly applied marking paint.
- b. The method or product shall not damage the existing pavement, shoulder or natural landscape.
- c. This method or product should be fully described in the Paint Spill Cleanup Plan, for proper review and approval.
- d. The plan should anticipate possible paint or solvent spills on various substrates such as asphalt pavement, grass shoulders or natural landscapes within the work site of this project.

PART 3 – WORK REQUIREMENTS

3-1 GENERAL

- A. The contractor is responsible for investigating site conditions in the vicinity of the work including frost depths, groundwater, location of rock, existing utilities, or any other hazards or obstacles which may increase time or cost of the work.
- B. Contracts for other construction projects may be underway or may commence during the construction of this project. Coordinate with the COR and other contractors to minimize disruption to this project and to other construction projects in the area.

3-2 SITE SURVEYING AND CONSTRUCTION STAKING

- A. The intent of this contract is to match existing striping such as double solid, single solid, dashed, or skips.
- B. If new striping is required, coordinate with the COR for the layout prior to beginning work.

3-3 TRAFFIC CONTROL

- A. Temporary Traffic Control Devices (TTC) shall be installed and maintained in accordance with the current version of the Manual on Uniform Traffic Control Devices (MUTCD) and the approved Safety and Traffic Control Plan. TTC shall be covered or removed during periods of no work.
- B. Furnish sufficient quantity of personnel and equipment to operate a mobile operation for the work of this project.
- C. If operations have to be temporarily stationary such as near an intersection, provide a flagger operation. Limit traffic delays to a maximum of 10 minutes.
- D. Keep traffic on the paved roadway (do not use the road shoulder to move traffic past the striping operation).
- E. All Contractor vehicles shall be equipped with white or amber strobe lights or rotating beacons installed atop the vehicles. These warning lights shall operate at all times when vehicles are located at the job site, unless the vehicle is not in use and parked well away from the active traffic lanes.
- F. Personnel working on the job shall have and use required personal protective equipment such as high visibility safety vests and hard hats.
- G. National Park Service Law Enforcement will not be available to assist with traffic control.
- H. Refer to SOW Section 1-3.B.3 for qualified personnel requirements.

3-4 WORK EXECUTION

A. Layout of work

1. The intent of this project is to restripe existing, faded pavement markings. It is anticipated that little to no layout work will be required.
2. Before placing any striping paint, mark badly faded existing pavement markings as necessary by placing spots of paint at start and end points and at intervals to ensure accurate location of lines on the road.

B. Roadway Surface Preparation

1. Prepare the surface for striping in accordance with manufacturer specifications.
2. Prior to applying pavement markings, the Contractor shall clean the pavement to be marked by brooming, airblasting, high pressure water, manual scrubbing with water and approved detergent, or other acceptable cleaning method.
3. Surface must be dry, free of dust, dirt, leaves, loose gravel, oil, grease, debris and other foreign contaminants before painting.

C. Roadway Striping Application

1. General

- a. Match existing colors, patterns, and spacing. Advise the COR of any locations which do not meet the MUTCD and coordinate with the COR a resolution for these locations.
- b. The installed pavement marking material shall have a uniform thickness and smooth surfaced cross section throughout its entire length, while maintaining the listed tolerances and application rates.
- c. Centerlines shall be 4-inch-wide stripes; yellow paint with reflective beads. Separate double centerline stripes with 4-inch space.
- d. The Contractor shall protect the pavement markings until track free.
- e. Any markings tracked by a vehicle shall be removed by methods acceptable to the COR and at no additional expense to the Government.
- f. The Contractor shall be responsible for removing all roadway striping materials spilled on the road surface by a method acceptable to the COR.

2. Tolerances

- a. New striping shall match to the greatest extent possible the of the existing striping.
- b. Stripes shall begin and end within 2 inches of the ends of the existing markings.
- c. Any greater deviations shall be sufficient cause for requiring the material to be removed and repainted at no additional cost to the Government.
- d. All stripes shall be plus or minus 1/4 inch of the required stripe width and length.

3. Application Rates

- a. All longitudinal pavement stripes 8 inches or less in width shall be applied with one pass of the pavement marking equipment.
- b. Stripes greater than 8 inches in width and pavement marking symbols may be applied with multiple passes of the pavement marking equipment.

- c. Spray paint at 25 mil (0.63 millimeters) minimum wet film thickness or at a rate of 71 square feet per gallon (1.7 square meters per liter).
 - 4. Glass Bead Application
 - a. Use a drop-on method of application, where the glass beads are dispensed by a pressurized mechanical feed or a high pressurized means onto the pavement marking as it is applied to the pavement. Drop-on bead dispensing methods for pavement marking lines shall not be simply dependent on gravity or dropped by hand.
 - b. Approximately 70 percent of the beads shall be fully buried in paint with the remaining 30 percent embedded 50 percent into the paint surface.
 - c. Use two bead dispensers. Apply Type III glass beads on the paint at a rate of 6 to 8 pounds per gallon (0.72 to 0.960 kilograms per liter) followed by Type I glass beads on the paint at a rate of 6 to 8 pounds per gallon (0.72 to 0.96 kilograms per liter) of paint.
- D. Repair of Roadway Striping
 - 1. The Contractor shall be responsible for removing and replacing any improperly applied, smeared, tracked or damaged pavement markings at no additional cost to the Government.
 - 2. Improperly applied and damaged markings shall be removed using methods that will not damage the pavement surface or leave confusing patterns on the pavement similar to the traffic markings being removed.
 - 3. Methods used for paint removal shall be detailed in the Paint Spill Cleanup Plan.
 - 4. Remove any excess material resulting from the marking removal and replacement pavement markings.
- E. Paint Spill Cleanup
 - 1. Prevention
 - a. Contractor shall take precautions to prevent spills when transporting paint, filling striping truck paint reservoirs, and cleaning paint nozzles on the equipment.
 - b. Whenever possible, move equipment off of the pavement during filling and cleaning operations.
 - c. Use material such as building paper, plastic sheeting, drop cloths, or other disposable sheeting to protect the ground and/or pavement surface from paint spills.
 - 2. Spill Cleanup
 - a. Paint spills shall be immediately cleaned up at the time of occurrence.
 - b. The crew shall have on hand at all times the necessary materials and equipment to perform effective spill cleanups.
 - c. If spills occur on road shoulders or other non-paved areas, remove and dispose of any contaminated soil and/or vegetation.
 - d. Methods and materials used for paint spill cleanup shall be those approved as part of the Paint Spill Cleanup Plan.

3. Paint on Vehicles

- a. It is the responsibility of the Contractor to protect the pavement markings until dry.
- b. The Contractor is responsible for resolving issues with paint getting onto visitor vehicles.

F. Roadway Striping Protection

- 1. Protect markings from traffic until dried to a no-tracking condition. Remove tracking marks, spilled marking material, markings in unauthorized areas, and defective markings.
 - a. Place traffic cones on all freshly painted traffic markings to keep traffic off wet markings and prevent smearing, tracking or movement of the markings.
 - i. Curves: Place cones at intervals of 50 foot or less in curved road sections.
 - ii. Straight Sections: Place cones at intervals of 100 feet or less on all straight or nearly straight road sections.
 - iii. Intersections: Place cones at ends and sides of markings.
 - iv. On-Ramps: Place cones at intervals of 25 feet or less at on-ramps which have a tight turning radius.
 - v. Parking Lots: For small parking pull offs, place traffic cones or suitable barricades to close off each entrance to the parking lot until striping is placed and dry. On larger parking lots, place cones at locations to protect the sections of the parking lot striping being installed, or rope off areas, until the striping is dry.
 - b. Remove all cones as soon as paint has dried or cured sufficiently to prevent smearing, tracking or movement of the markings.
- 2. For centerline striping, the distance between the cone placement vehicle and the paint striping truck shall be maintained at no more than 150 feet at all times.
 - a. The cone pickup vehicle shall travel with the traffic flow and no part of the vehicle shall cross or straddle the centerline.
 - b. When five or more vehicles accumulate behind the cone pickup vehicle, the cone pickup vehicle should exit the roadway at the first convenient and safe location and allow traffic to pass.
 - c. The cone pickup vehicle shall also allow following traffic to pass at intervals not to exceed 5 minutes.

3-5 CLEAN UP

- A. Routine Cleanup: Keep project site neat, orderly and in safe condition at all times. Provide enough refuse containers for collecting construction debris. Pick up construction debris daily to prevent blowing and scattering. Keep volatile wastes in covered containers. Unless otherwise specified, all removed material becomes the property of the contractor and shall be disposed of outside of the park in accordance to local, state and county regulations.
- B. Final Cleanup: Remove all tools, equipment, surplus materials and rubbish from the site upon completion of the work. Restore or refinish surfaces of existing facilities that are marred, scratched or damaged due to work of this contract, to match the original condition. Remove any grease, dirt, asphalt, stains or foreign material from the finished surfaces. Use cleaning materials recommended

by the manufacturer of the surface to be cleaned. Sweep paved or concrete areas and rake grounds to remove all construction debris. At time of the final inspection, project site shall be thoroughly clean and ready for use. All nails shall be found, collected and removed during the final cleanup of the site and through the entirety of the project at the end of each workday.

3-6 CLOSEOUT

- A. Closeout Submittals: Upon completion of the work, the Contractor shall provide any final submittals called for in SOW PART 2.
- B. Substantial Completion and Final Inspection: Submit written certification that project, or designated portion of the project, is substantially complete, and request in writing a final inspection. The COR or designated representative will inspect within 10 days of receipt of request. When the COR determines that the work is substantially complete, it will prepare a list of any deficiencies to be corrected before final acceptance and issue a Letter of Substantial Completion. If the COR determines the work is not substantially complete, the Contractor will be notified in writing, stating reasons. After completing work, Contractor shall resubmit certification and request a new final inspection.

PART 4 – MEASUREMENT AND PAYMENT

4-1 GENERAL

Provide all supervision, labor, materials, transportation, equipment, tools and incidentals to complete execution as specific in this SOW and related references.

4-2 MEASUREMENT

This is a lump sum contract. Individual items of work will not be measured for payment. Quantities are provided for estimating and planning purposes.

4-3 PAYMENT

This a lump sum contract with one final payment made at the completion and acceptance of the project. If the Contractor wishes to receive partial payments, a proposed payment schedule for the project work shall be submitted to the CO for review and approval. Partial payments can then be made providing all necessary documentation (e.g. payrolls, daily reports, photos, etc.) has been submitted to support the partial payment request/invoice.

END OF STATEMENT

ATTACHMENTS